

Developing a domain-specific LLM through self-instruct fine-tuning

Application to reliability engineering

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Outline

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Reliability engineering challenges

- Complex systems: nuclear plants, aircraft, electrical grids
- Requires sophisticated mathematical modeling
- Significant manual programming effort
- Specialized domain knowledge needed

General LLMs fall short

- Built for general programming/tasks
- Poor performance on reliability-specific tasks
- Lack of specialized training data
- Cannot reason through multi-step domain problems

Key issue

State-of-the-art LLMs consistently provide **incorrect or nonsensical code** for reliability engineering tasks due to lack of domain exposure during pre-training.

- Previous students created **58 hand-written reliability coding tasks** with unit tests
- Evaluated state-of-the-art LLMs on this benchmark
- Results were consistently poor across all models

Common failure modes:

- Lack of domain specific formulas
- Hallucinated reasoning steps/formulas

Project objectives

Primary goal

Develop a highly effective, domain-specific LLM (reasonably sized) for reliability engineering through fine-tuning with synthetic data

Three key components: maybe rework this:

① **Synthetic data generation**

Use self-instruct strategy to automatically generate large corpus of Q&A pairs

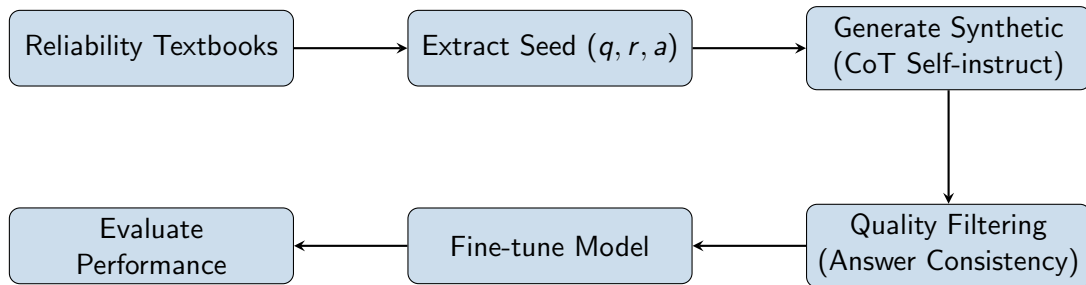
② **Chain-of-Thought integration**

Generate step-by-step reasoning for complex reliability problems

③ **Model fine-tuning & Evaluation**

Fine-tune open weight models to have domain knowledge and evaluate performance

Methodology Overview



Self-Instruct & Synthetic Data Generation

Self-Instruct (Wang et al., 2023)

Pioneering framework for bootstrapping instruction-following capabilities from LLMs themselves. Uses seed demonstrations to generate diverse synthetic instruction-output pairs, achieving 33% improvement over vanilla GPT-3 on Super-NaturalInstructions.

CoT-Self-Instruct (Yu et al., 2024)

Extends self-instruct by incorporating Chain-of-Thought reasoning during generation. Uses Answer-Consistency filtering for verifiable tasks. Outperforms original 'seed' datasets (s1k, OpenMathReasoning) on MATH500, AMC23, AIME24, and GPQA-Diamond benchmarks.

Relevance to our work: We adopt the self-instruct paradigm with consistency-based quality filtering to generate synthetic reliability engineering questions from textbook seeds.

Chain-of-Thought Reasoning

Auto-CoT (Zhang et al., 2022)

Eliminates manual effort in creating CoT demonstrations by using diversity-based question clustering and automatic reasoning chain generation.

Key innovation:

- Cluster questions by similarity
- Generate reasoning chains with "Let's think step by step"
- Mitigate errors through diversity

Self-Refine IT (Ranaldi & Freitas, 2024)

Two-stage alignment: (1) Transfer reasoning via instruction-tuning on LLM demonstrations, (2) Self-refinement through Direct Preference Optimization (DPO). Enables smaller models to learn step-wise reasoning without separate reward models.

Results:

- Outperforms standard instruction-tuning
- Strong in/out-domain generalization
- Preference optimization for reasoning

Implication: CoT and preference-based optimization could be interesting next steps.

Two complementary question types

Exercise-based questions

- Numerical reliability problems
- Multi-step mathematical derivations
- Code-oriented (MTTF, simulations, distributions)

Derive the Mean Time To Failure (MTTF) for:

- a 2/3 voting system
- a 3/5 voting system

Assume negligible voter failure probability.

Answer: 5/6, 47/60

Theory-based questions

- Definitions, assumptions, interpretations
- “Why / How” reasoning
- Reliability theory understanding

Which of the following is Bayes Theorem for 2 events A, B ?

- A) $P(A|B) = P(B|A)P(A)/P(B)$
- B) $P(A|B) = P(A \cap B)/P(B)$
- C) ...

Answer: A

Step 1: Question/Answer extraction

- Scan textbooks for keywords such as (Exercise/Problem + Solution) with regex
- Extract all surrounding text

Step 2: Quality validation with SLM

- Remove samples that have or refer to figure/table references
- Validate self-containment and whether it meets our question requirements
- Verify clear, unambiguous answers

Step 3: Reformatting data with SLM

- Separate text into 3 components using an LLM:
- Question
- Reasoning
- Answer

Result

288 high-quality seed (q, r, a) samples for baseline evaluation and synthetic generation

Synthetic data generation

Self-instruct strategy:

- 1 Sample 2 seed questions randomly
- 2 Prompt LLM to create novel inspired question
- 3 Generate step-by-step solution with CoT
- 4 Extract new (q, r, a)

Quality control - answer consistency checking:

- Re-solve generated question n times (with somewhat high temperature)
- Calculate consistency score: matches / total attempts
- Accept only both the score ≥ 0.5 and the majority prediction is equal to initial answer
- Filters out ambiguous or incorrectly formulated questions

Generation Results

Generated **81 synthetic questions** that passed consistency filtering

Baseline evaluation results

Exercise-based Q&A

Model	Dataset	Total	Correct	Accuracy
GPT-4-Mini	Seed	288	122	42.4%
Qwen3-14B	Seed	288	100	34.7%
GPT-4-Mini	Synthetic	81	74	91.4%
R1-Distill-Qwen3-14B	Synthetic	81	77	95.1%

Conceptual Q&A (MCQs)

Model	Total	Correct	Accuracy
GPT-4.1-Mini	231	219	95.0%
R1-Distill-Qwen3-14B	231	214	92.7%

→ Conceptual questions already achieve near-ceiling performance, leaving limited room for improvement through fine-tuning. As a result, we focus fine-tuning efforts on exercise-based questions, where significant performance gains are still possible.

Baseline evaluation results

Model	Dataset	Correct	Total	Accuracy
GPT-4-Mini	Seed	122	288	42.4%
R1-Distill-Qwen3-14B	Seed	100	288	34.7%
GPT-4-Mini	Synthetic	74	81	91.4%
R1-Distill-Qwen3-14B	Synthetic	77	81	95.1%

Key observation

Synthetic questions have **much higher** accuracy which makes sense as a similar capacity model created these questions

Open questions

Does this matter for finetuning? Does a smaller LM even have the capacity to learn questions generated by larger LM?

Fine-tuning setup & infrastructure

- Fine-tuning experiments conducted on **La Ruche** (CentraleSupélec HPC cluster)
- Dedicated training session on large-scale model fine-tuning
- Guidance on:
 - GPU job scheduling and resource allocation
 - LoRA-based fine-tuning workflows
 - Experiment monitoring and debugging
- Training provided by **Solal** (CentraleSupélec)

Fine-tuning results

Fine-tuning setup:

- Model: Qwen3-14B with LoRA adapters
- Training data: 230 Seed Q&A pairs

Results for held out Q&A pairs

Model	Correct	Total	Accuracy	Change
Unfinetuned Qwen	5	58	8.6%	—
Finetuned Qwen	8	58	13.8%	+5.2%

Result

60% relative improvement in accuracy on seed questions shows that the models are able to learn how to solve these problems and finetuning allows the models to better generalize

Challenges encountered

① Data quality gap

- A good fine-tuning requires CoT and not only direct answers
- Models perform well on generated data but poorly on textbook questions
- Need better generation strategies to match complexity

② Limited training data

- Only 81 synthetic questions after filtering
- Original goal: thousands of examples
- Consistency filtering too strict? (50% threshold)

③ Domain complexity

- Reliability engineering requires deep statistical knowledge
- Requires human validation of generated QAs
- Multi-step reasoning with specialized distributions
- Numerical precision matters (small errors = incorrect)

Future directions

Data generation

- Increase synthetic dataset size (500+)
- Use textbook context as seed material
- Improve consistency checking
- Generate chain-of-thought solutions
- Add difficulty stratification

Model training

- Full fine-tuning (not just LoRA)
- Multi-stage training pipeline
- Reinforcement learning from feedback

Evaluation

- Expand test set beyond 58 questions
- Human expert evaluation
- Code execution testing
- Real-world problem benchmarks

System Integration

- Interactive coding assistant
- Auto-correction with verification
- Uncertainty quantification
- Explainable AI for solutions

Domain-specific application: Financial Reasoning

Fin-R1 (Liu et al., 2025)

Domain-specific reasoning LLM for finance using two-stage training: (1) SFT on 60K CoT financial examples distilled from DeepSeek-R1, (2) RL with GRPO (Group Relative Policy Optimization) using format and accuracy rewards. Achieves 7B parameter model with SOTA performance on FinQA and ConvFinQA, rivaling much larger models.

Methodology Parallels:

- Domain-specific dataset curation
- Quality filtering (LLM-as-judge)
- Parameter-efficient training
- Numerical reasoning focus

Key Differences:

- Finance: 60K examples vs our 81
- Uses RL phase (we use only SFT)
- Distills from stronger teacher model
- Structured output format enforcement

Lesson: Domain-specific LLMs require substantially larger datasets (1000s, not 10s) and benefit from RL-based refinement for complex reasoning tasks.

Key findings

- ❶ **Problem confirmed:** General LLMs perform poorly on reliability engineering tasks (8.6% baseline accuracy)
- ❷ **Fine-tuning works:** Domain-specific training improves performance by 60% relative improvement (8.6% $\rightarrow$ 13.8%)
- ❸ **Data quality critical:** Synthetic questions much easier than real textbook problems (95% vs 14% accuracy)
- ❹ **Scale matters:** Current dataset too small (81 questions) for robust domain adaptation
- ❺ **Relevance of CoT:** Learning with reasoning steps is essential; otherwise, models learn to reproduce answers without learning the underlying problem-solving process.

Main conclusion

Self-instruct fine-tuning is a **viable approach** for domain-specific LLM development, but requires **larger, higher-quality** training datasets that better match real-world problem complexity.

Thank you for your attention!

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Slides and code available on GitHub

References & Resources I

Key Papers:

- ① Wang, Y. et al. (2023). *Self-Instruct: Aligning Language Models with Self-Generated Instructions*. ACL 2023.
- ② Zhang, Z. et al. (2022). *Automatic Chain of Thought Prompting in Large Language Models*. ICLR 2023. arXiv:2210.03493.
- ③ Ding, B. et al. (2024). *Data Augmentation using Large Language Models: Data Perspectives, Learning Paradigms and Challenges*. ACL 2024 Findings. arXiv:2403.02990.
- ④ Ranaldi, L. & Freitas, A. (2024). *Self-Refine Instruction-Tuning for Aligning Reasoning in Language Models*. EMNLP 2024. arXiv:2405.00402.
- ⑤ Yu, P. et al. (2024). *CoT-Self-Instruct: Building High-Quality Synthetic Data for Reasoning and Non-Reasoning Tasks*. arXiv:2507.23751.
- ⑥ Liu, Z. et al. (2025). *Fin-R1: A Large Language Model for Financial Reasoning through Reinforcement Learning*. arXiv:2503.16252.

Models Used:

- Qwen3-14B (Alibaba Cloud)
- GPT-4-Mini (OpenAI)
- Mistral-Small-3.1-24B (Mistral AI)

Tools & Libraries:

- OpenRouter API
- Python (extractors, evaluators, generators)
- LoRA (Low-Rank Adaptation)